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# =============================
# Titanic Data Cleaning & Preprocessing
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# Import libraries
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.preprocessing import LabelEncoder, StandardScaler

# -----------------------------
# 1. Load Dataset
# -----------------------------
df = pd.read_csv("Titanic-Dataset.csv")

# -----------------------------
# 2. Display Basic Information
# -----------------------------
print("First 5 Rows:")
print(df.head())

print("\nDataset Info:")
print(df.info())

print("\nMissing Values:")
print(df.isnull().sum())

# -----------------------------
# 3. Handle Missing Values
# -----------------------------

# Fill Age with median
df['Age'].fillna(df['Age'].median(), inplace=True)

# Fill Embarked with mode
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)

# Fill Cabin with Unknown
df['Cabin'].fillna('Unknown', inplace=True)

# Check missing values again
print("\nMissing Values After Cleaning:")
print(df.isnull().sum())

# -----------------------------
# 4. Encode Categorical Columns
# -----------------------------
label_encoder = LabelEncoder()

df['Sex'] = label_encoder.fit_transform(df['Sex'])
df['Embarked'] = label_encoder.fit_transform(df['Embarked'])

# -----------------------------
# 5. Drop Unnecessary Columns
# -----------------------------
df.drop(['Name', 'Ticket', 'Cabin'], axis=1, inplace=True)

# -----------------------------
# 6. Detect Outliers
# -----------------------------
plt.figure(figsize=(10,5))
sns.boxplot(data=df[['Age', 'Fare']])
plt.title("Boxplot Before Outlier Removal")
plt.show()

# -----------------------------
# 7. Remove Outliers (IQR Method)
# -----------------------------
Q1 = df['Fare'].quantile(0.25)
Q3 = df['Fare'].quantile(0.75)

IQR = Q3 - Q1
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lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

df = df[(df['Fare'] >= lower) & (df['Fare'] <= upper)]

# -----------------------------
# 8. Feature Scaling
# -----------------------------
scaler = StandardScaler()

columns = ['Age', 'Fare', 'SibSp', 'Parch']

df[columns] = scaler.fit_transform(df[columns])

# -----------------------------
# 9. Final Dataset
# -----------------------------
print("\nProcessed Dataset:")
print(df.head())

print("\nShape After Preprocessing:")
print(df.shape)

# -----------------------------
# 10. Save Cleaned Dataset
# -----------------------------
df.to_csv("Titanic_Cleaned.csv", index=False)

print("\nData Cleaning & Preprocessing Completed Successfully!")
```